

KARLSRUHE BADEN BADEN, Germany

WMO: 107275

Lat: 48.779N

Lon: 8.081E

Elev: 124

StdP: 99.84

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-7.1	-5.1	-11.2	1.5	-5.0	-8.1	1.9	-3.3	12.2	9.1	11.0	5.4	2.9	30	0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	32.2	20.6	30.1	20.1	28.1	19.2	21.7	29.3	20.9	28.0	20.1	26.5	3.1	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.2	24.0	18.3	13.4	23.3	17.9	13.1	23.0	64.0	29.3	61.1	28.2	58.4	26.5	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 10.0	8.8	7.7														
(5)			DB	-10.5	36.1	3.6	1.4	-13.1	37.1	-15.2	37.9	-17.3	38.7	-19.9	39.7	
			WB	-10.8	23.8	3.6	1.0	-13.4	24.5	-15.5	25.1	-17.5	25.6	-20.1	26.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	2.5	3.0	6.9	11.4	15.2	19.0	20.9	20.0	16.1	11.4	6.4	3.5
	DBStd	7.54	4.37	4.37	4.07	3.71	3.64	3.47	3.16	3.27	3.06	3.69	3.69	3.99
	HDD10.0	927	235	197	111	28	3	0	0	0	1	29	119	203
	HDD18.3	2774	492	429	356	208	111	32	12	19	82	217	358	460
	CDD10.0	1435	2	2	13	70	163	271	337	311	182	72	11	2
	CDD18.3	242	0	0	0	1	12	54	91	71	13	1	0	0
	CDH23.3	2312	0	0	0	30	145	509	852	642	128	6	0	0
	CDH26.7	778	0	0	0	3	29	164	325	232	25	0	0	0
(14) Wind	WSAvg	3.3	4.0	3.7	3.8	3.2	3.2	2.9	3.0	2.7	2.8	3.0	3.2	3.8
(15) Precipitation	PrecAvg	931	75	69	68	60	87	83	89	77	69	81	85	89
	PrecMax	1180	160	149	175	119	189	181	201	152	129	224	167	214
	PrecMin	676	9	17	16	2	26	29	31	11	25	11	5	12
	PrecStd	138	35	37	43	30	38	35	38	34	29	44	37	49
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	15.2	20.9	26.2	30.1	33.8	35.2	35.2	29.2	23.9	17.9	14.1
		MCWB	10.3	9.3	11.8	15.0	19.3	20.9	21.4	21.2	18.6	17.5	13.6	10.6
	2%	DB	12.2	12.9	18.2	23.2	27.0	31.0	32.8	31.8	26.9	20.9	15.6	12.2
		MCWB	9.4	8.7	10.6	13.9	17.5	20.1	20.7	20.6	18.8	16.1	12.2	10.0
	5%	DB	11.0	11.0	16.0	21.1	24.9	28.8	30.2	29.1	24.2	18.8	13.9	10.9
		MCWB	8.5	7.7	9.8	13.3	16.4	19.7	19.9	20.0	17.7	14.9	11.3	8.8
	10%	DB	9.2	9.1	13.8	19.0	22.6	26.2	28.0	26.9	22.1	16.9	12.1	9.2
		MCWB	7.3	6.7	9.3	12.2	15.5	18.6	19.0	19.2	16.6	13.9	9.9	7.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.3	10.5	12.8	16.2	20.9	22.9	22.6	22.8	20.6	17.7	14.2	11.8
		MCDB	13.4	13.7	18.6	24.1	27.6	30.1	31.3	30.2	26.6	21.9	16.7	13.3
	2%	WB	10.0	9.2	11.7	14.7	19.2	21.2	21.5	21.6	19.4	16.4	12.6	10.4
		MCDB	12.1	12.3	16.0	21.7	25.2	28.4	29.5	28.9	24.9	19.6	14.8	12.1
	5%	WB	8.7	8.0	10.7	13.6	17.5	20.3	20.8	20.7	18.2	15.3	11.4	9.0
		MCDB	10.8	10.5	14.7	19.9	22.6	26.8	27.9	27.5	23.0	18.2	13.7	10.7
	10%	WB	7.3	6.7	9.6	12.6	16.2	19.3	19.9	19.7	17.1	14.3	10.2	7.7
		MCDB	8.8	8.7	13.4	17.9	20.9	25.5	26.2	25.6	21.2	16.5	11.9	9.1
(35) Mean Daily Temperature Range	5% DB	MDBR	5.2	6.8	9.7	11.0	10.6	10.8	11.1	11.1	10.6	8.6	5.5	5.0
		MCDBR	6.9	9.9	14.4	16.0	14.9	14.4	15.7	15.0	14.1	11.4	7.9	6.7
		MCWBR	5.2	6.5	8.2	7.8	6.7	6.1	5.6	5.9	6.9	6.9	5.3	5.2
	5% WB	MCDBR	6.2	8.7	11.7	14.0	12.6	13.1	13.3	13.7	12.3	10.4	7.3	6.4
		MCWBR	5.0	6.2	7.0	7.0	6.2	5.8	5.2	5.9	6.5	6.7	5.3	5.1
(40) Clear-Sky Solar Irradiance	taub		0.322	0.347	0.379	0.412	0.422	0.427	0.424	0.421	0.394	0.373	0.342	0.319
	taud		2.424	2.355	2.286	2.225	2.237	2.260	2.257	2.290	2.336	2.385	2.418	2.415
	Ebn at Noon		737	796	826	835	841	837	833	818	802	755	700	686
	Edh at Noon		66	88	110	129	133	131	130	120	104	85	66	59
(44) All-Sky Solar Radiation	RadAvg		0.92	1.68	2.87	4.27	5.04	5.75	5.38	4.68	3.50	2.10	1.06	0.73
	RadStd		0.13	0.28	0.43	0.59	0.55	0.47	0.51	0.46	0.38	0.20	0.12	0.12

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (72 neighbors)	N/A	N/A	N/A	N/A	+0.34	+0.27	N/A	N/A	+51	N/A

Nomenclature: See separate page